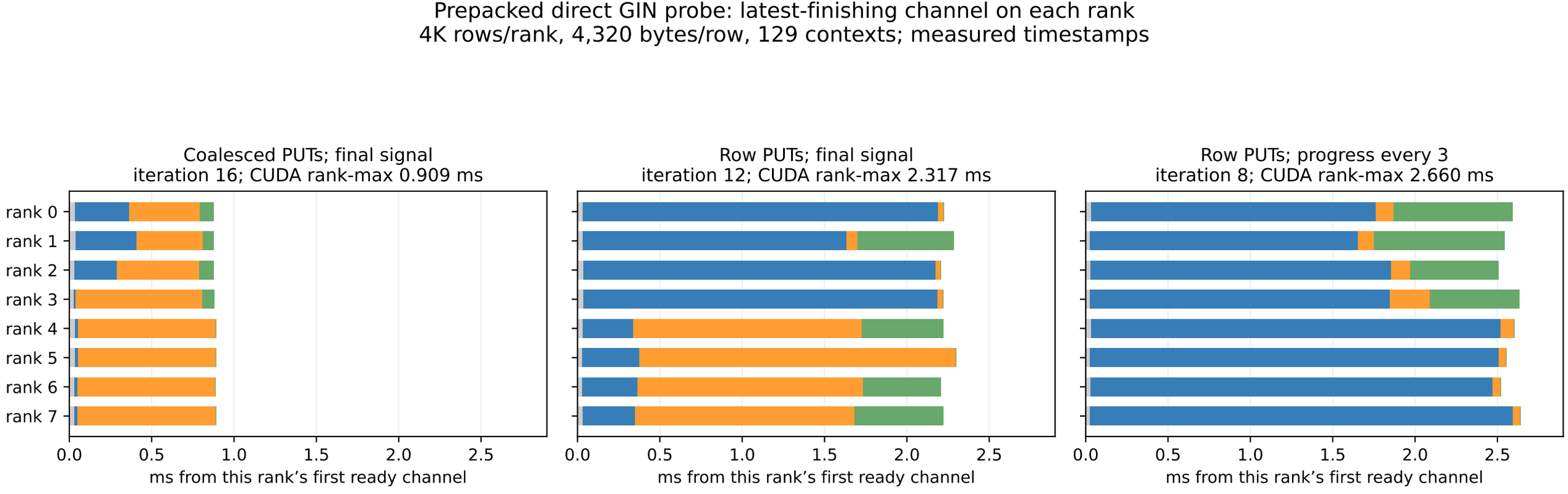
Prepacked direct GIN probe: latest-finishing channel on each rank
4K rows/rank, 4,320 bytes/row, 129 contexts; measured timestamps
Coalesced PUTs; final signal
iteration 16; CUDA rank-max 0.909 ms
Row PUTs; final signal
iteration 12; CUDA rank-max 2.317 ms
Row PUTs; progress every 3
iteration 8; CUDA rank-max 2.660 ms
rank 0
rank 1
rank 2
rank 3
rank 4
rank 5
rank 6
rank 7
0.0
0.5
1.0
1.5
2.0
2.5
ms from this rank's first ready channel
Delay until this channel is ready
Submit (includes periodic signaling)
Final signal + completion flush
Wait for remote readiness
Each row uses its own rank-local clock origin. Colored intervals are sequential on the selected channel; other channels overlap.
Completion includes transfer progress; it is not isolated NIC wire time. No NVLink relay, expert work, or initial readiness exchange is shown.